





**Department of Computer, Information Sciences, and Mathematics**

**Web Development I**

**Project Documentation**

# LK Bakery

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# Introduction

## 1.1 Rationale

### Background

In the age of the digital world, businesses can benefit greatly from an internet presence to connect and communicate efficiently with customers. LK Bakery, despite possessing a base of customers, is at a disadvantage without a dedicated website. Without a website, they have been at a loss in reaching potential clients beyond walk-in customers or messages on Facebook. Moreover, their lack of a website limits the ability to market their entire range of products and convey important details to customers. The website can also assist with maintaining a strong brand identity and attracting customer loyalty using digital media. These challenges pushed the bakery and the group into seeking a solution that would enhance communication, make products more visible, and develop customer connections.

### Business Opportunity

The website we have designed for LK Bakery includes several key elements that can help to solve the bakery's issues today. The Menu page allows customers to view the entire list of items, the Contact page allow easy access for customers for comments and contacting while giving location information for anyone interested visiting the bakeshop physically, and the About page also helps to set the brand image of the bakery given users a sense and idea of what the bakery is about and its upcoming. The following features are powered by a back-end system, which was implemented using PHP:

- Manage Menu: Use for product catalog management

- Manage Users: To view and manage user accounts for backend access
- Manage Categories: To manage and control the main and sub-categories of products. These traits allow the bakery to manage its operations better, thereby giving a better customer experience as well as easier internal control.

A primary objective of the site is to create an online presence for LK Bakery so that it can reach a wider range of customers efficiently. Aside from that, streamlining and creating processes for online connection and communication while building brand and loyalty are objectives that we seek to achieve with the implementation of the website.

## 1.2 Objectives

### 1.2.1. General Objectives

This project aimed to develop a website for LK Bakery in Cebu City, Philippines, over three months, featuring a customer-facing frontend and an administrative-focused backend that will strengthen online brand visibility, streamline operational workflows, and improve customer engagement for menu updates and user permissions.

### 1.2.2. Specific Objectives

**Objectives & Success Criteria**

- Reduce customer inquiries about menu details via a dynamic Menu page with item category, name, description, image, price, and availability, enabling informed purchasing decisions
- Improve accessibility through the Contact page's integrated map, opening hours, contact number, email, and message feature, ensuring seamless communication and reducing geographic barriers for local customers
- Increase brand loyalty using the About page to highlight LK Bakery's goal, history, and commitment to quality ingredients
- Decrease manual administrative tasks by enabling staff to add, edit, or remove items and categorize items via the Manage Menu and Category pages
- Secure backend operations with role-based permissions (admin, manager, staff, customer) to control access

**Business Risks**

- Inconsistent menu updates will harm customer trust
- Data breaches may expose user information
- Real-time inventory updates failure will cause overselling of out-of-stock items
- Staff resistance to backend tools slows operational efficiency gains
- Competitors launching similar features first will reduce LK Bakery's market differentiation

## 1.3. Beneficiaries

The proposed website will deliver targeted value to the following stakeholders:

**LK Bakery.** The website establishes a digital presence, broadening the bakery's reach beyond physical customers. Enhanced brand visibility through features like the About page and dynamic Menu page fosters customer loyalty. Backend tools streamline menu management and inventory updates, reducing manual tasks and improving operational efficiency.

**Administrators.** Admins oversee backend security, user permissions, and content management. Role-based access control minimizes data breach risks, ensuring secure operations. They maintain system health and optimize functionality, aligning with the project's security objectives.

**Managers.** Managers utilize backend tools to update product categories, monitor traffic analytics, and ensure accurate menu information. This supports data-driven decisions for inventory adjustments and marketing strategies, directly addressing business risks like inconsistent menu updates.

**Staff.** Staff members update daily inventory levels and categorize products via the backend, reducing manual paperwork and errors. This delegation improves workflow efficiency and ensures real-time product availability, mitigating overselling risks.

**Customers.** Customers gain 24/7 access to product details, pricing, store location (via integrated map), and contact options. The dynamic menu enables informed purchasing decisions, while the About page builds trust through transparency about the bakery's mission and history.

**Project Development Team.** The team acquires hands-on experience in full-stack development, including PHP backend integration, frontend design, and client communication.

This project enhances technical and collaborative skills, serving as a portfolio asset for academic and career advancement.

## 1.4. Scope and Limitations

This website focused on the three main variables: brand visibility, operational efficiency, and customer engagement of LK Bakery over a three-month period in Cebu City, Philippines. The capabilities and non-capabilities of the system are the following:

### Scope of Initial Release

- Centralized navigation to Home, About, Menu, and Contact pages
- Highlights the bakery's history, goal, and commitment to quality ingredients
- Displays product categories, names, descriptions, prices, and availability
- Integrated location map, opening hours, contact details, and a message form for inquiries
- Admin, Manager, and Staff roles with predefined permissions
- Manager and Staff can add, edit, or remove items
- Manager and Staff can sort items according to the main and sub categories

### Scope of Subsequent Releases

- Tracks customer behavior, sales trends, and campaign performance
- Rewards repeat customers with discounts or exclusive offers
- Staff training portal through an in-app tutorial to reduce resistance to backend tools

### Limitations and Exclusions

- Inventory updates require manual input by staff, as real-time synchronization with external delivery platforms is deferred to subsequent phases to prioritize foundational features

- The system does not support online payment processing, as LK Bakery currently prioritizes third-party platforms like Grab for transactions, avoiding redundancy and development complexity
- Integration with third-party delivery platforms via APIs is excluded from the initial release, as the focus remains on establishing brand visibility rather than order automation
- The website does not replace existing order channels and instead serves as a supplementary tool for brand storytelling and transparency
- Advanced features such as AI-driven recommendations or dynamic pricing are excluded to maintain simplicity and align with the bakery's immediate operational needs

## 1.5 Website Layout

### Wireframes

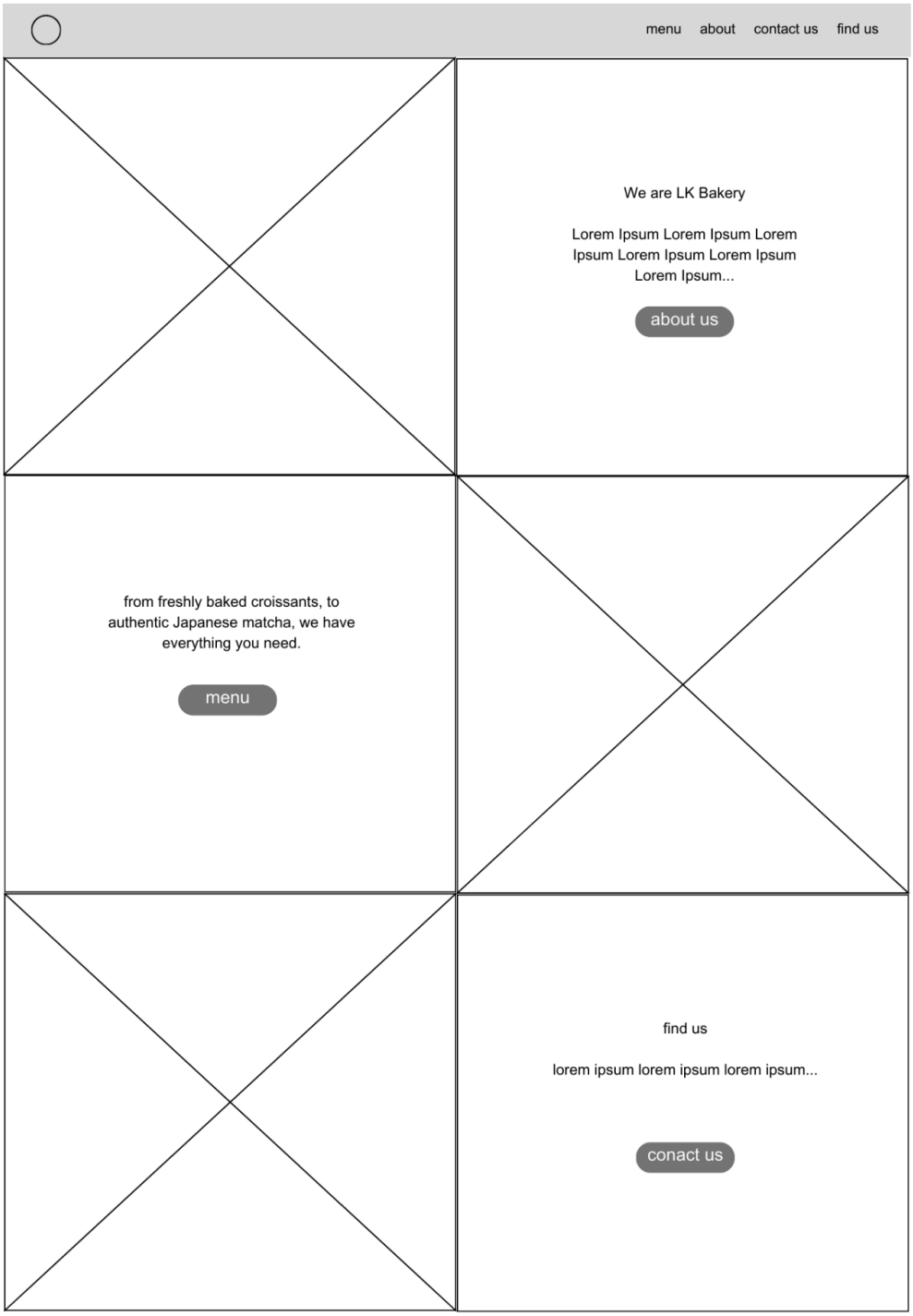



Figure 1.0 Home Page

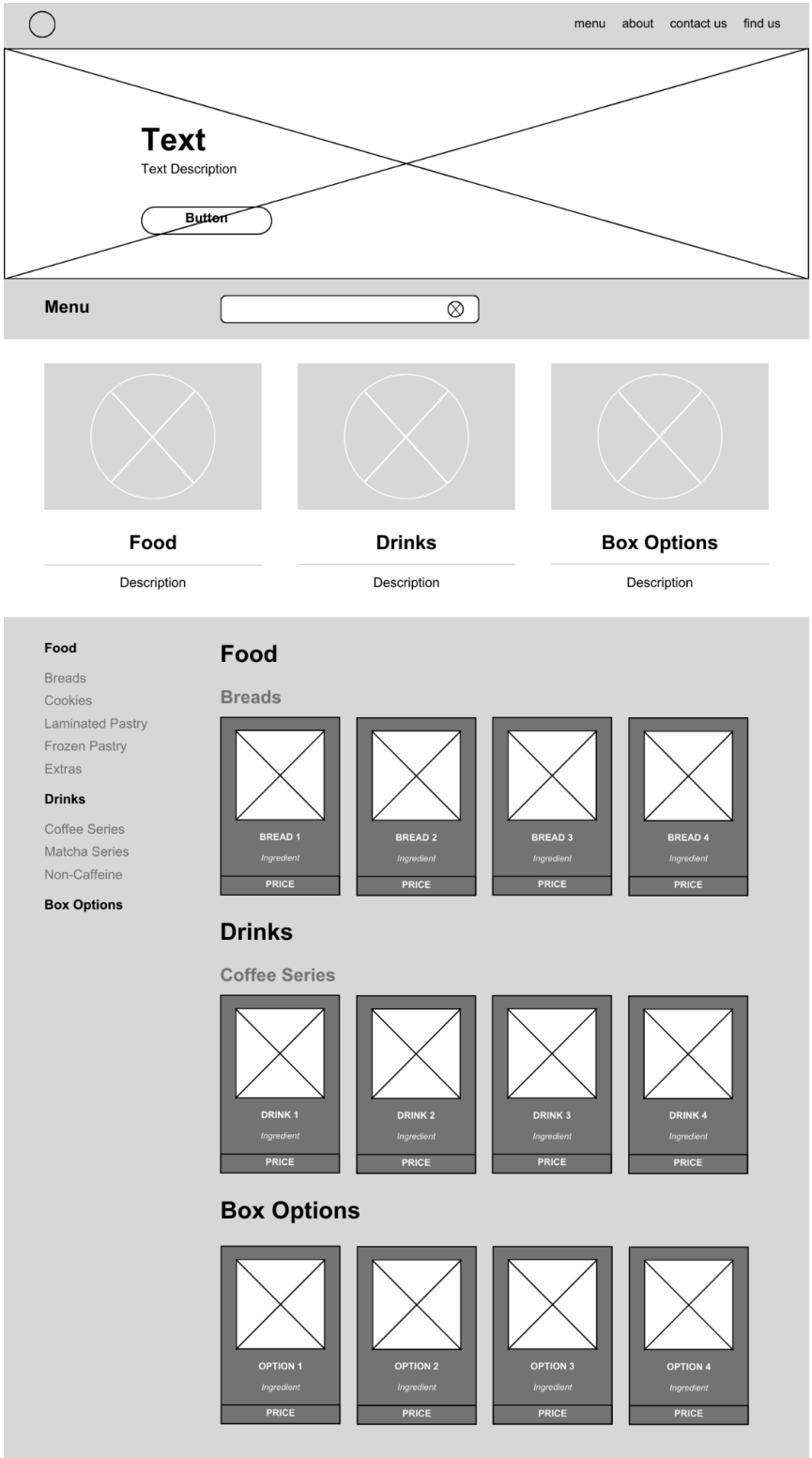



Figure 1.1 Menu Page

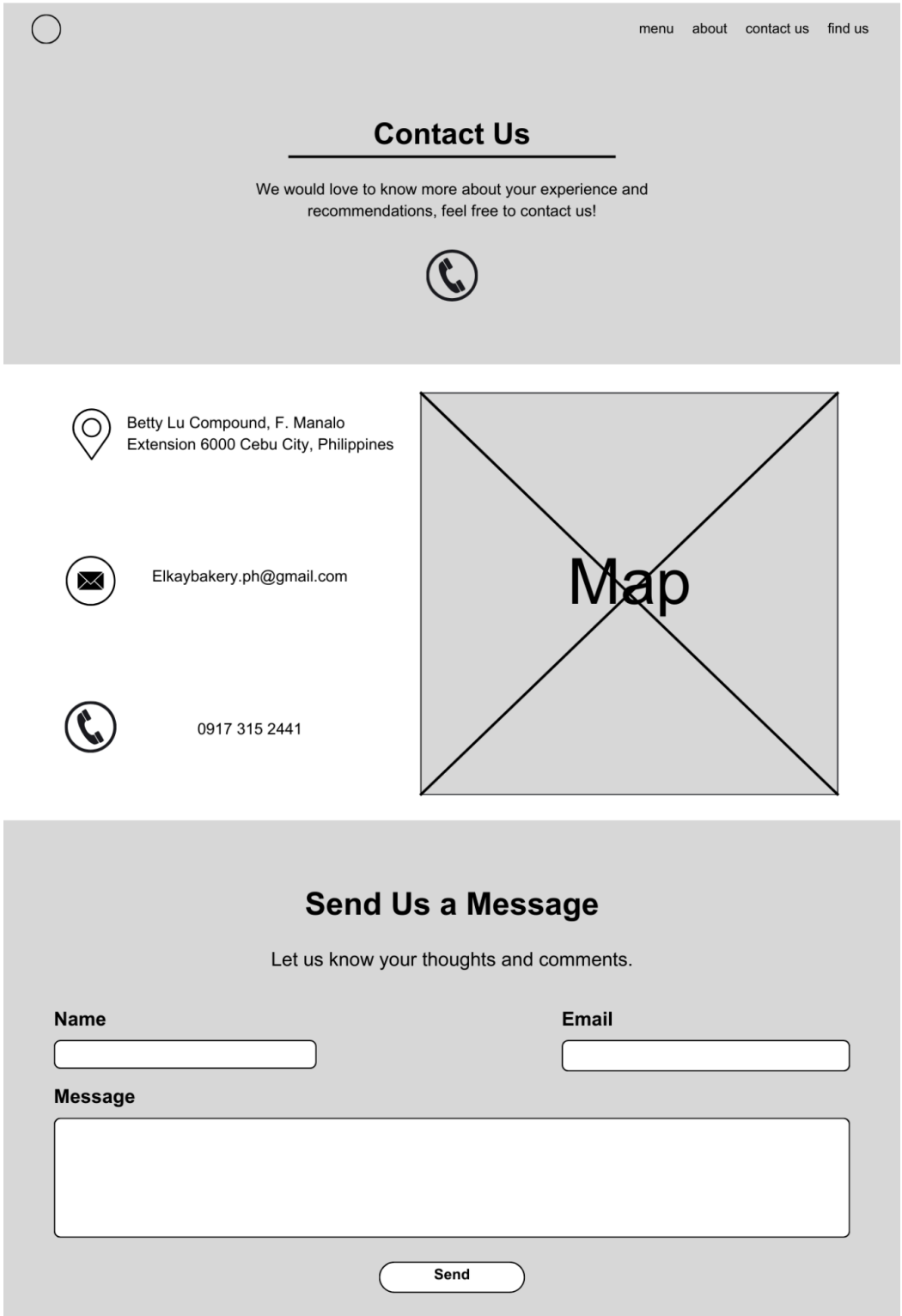



Figure 1.2 Contact Page

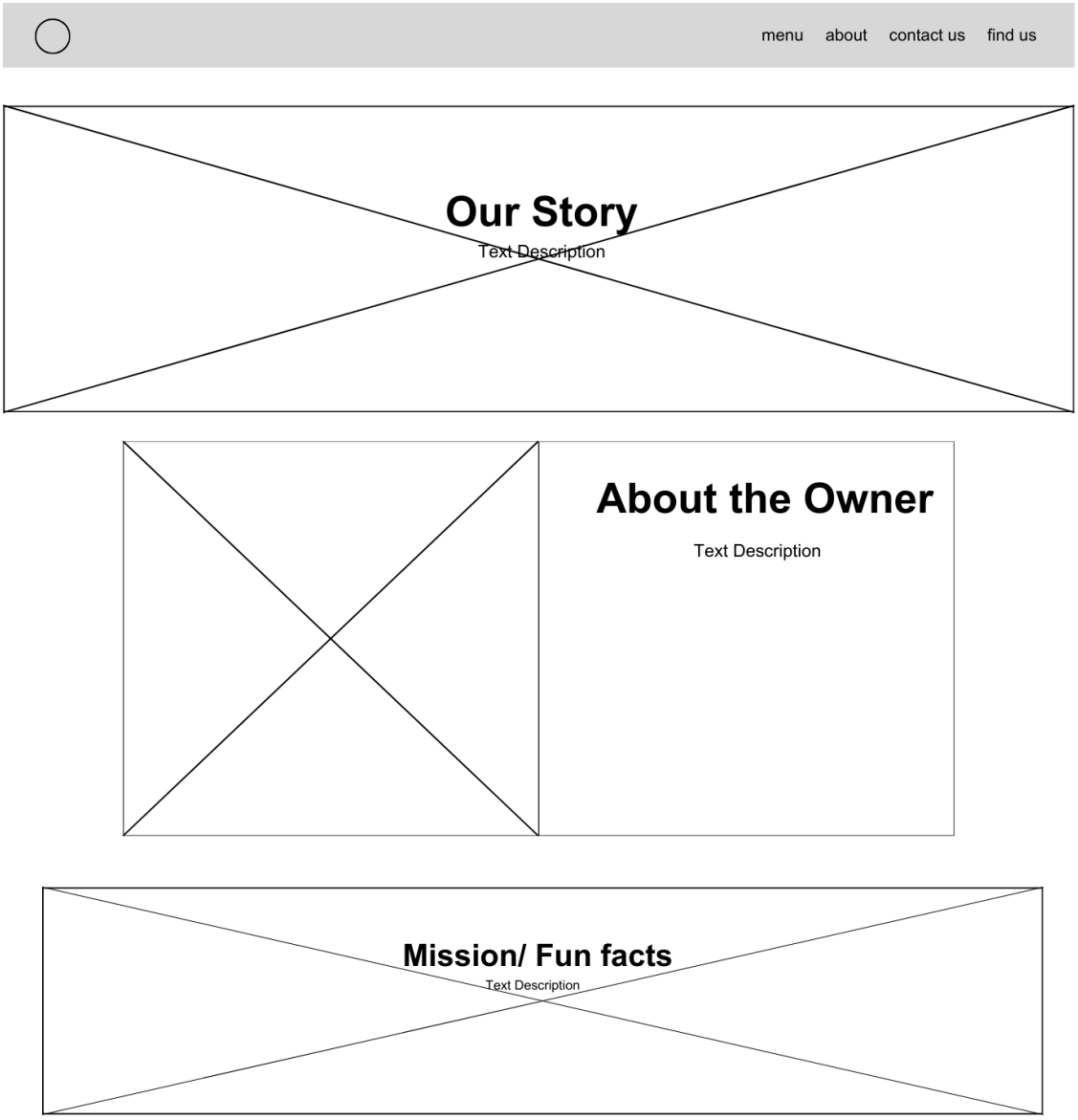



Figure 1.3 About Page

## User Interface

## Frontend

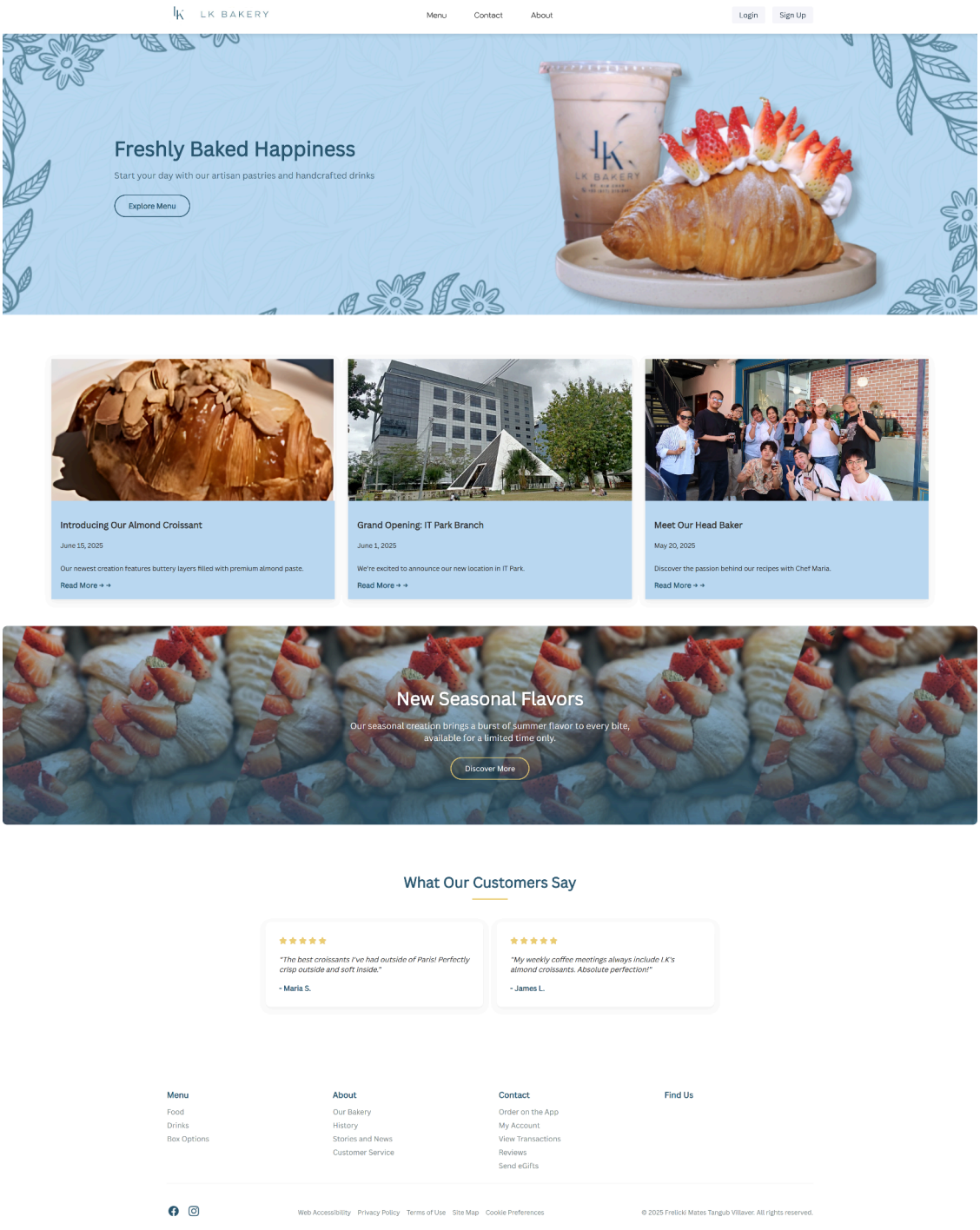



Figure 2.0 Home Page

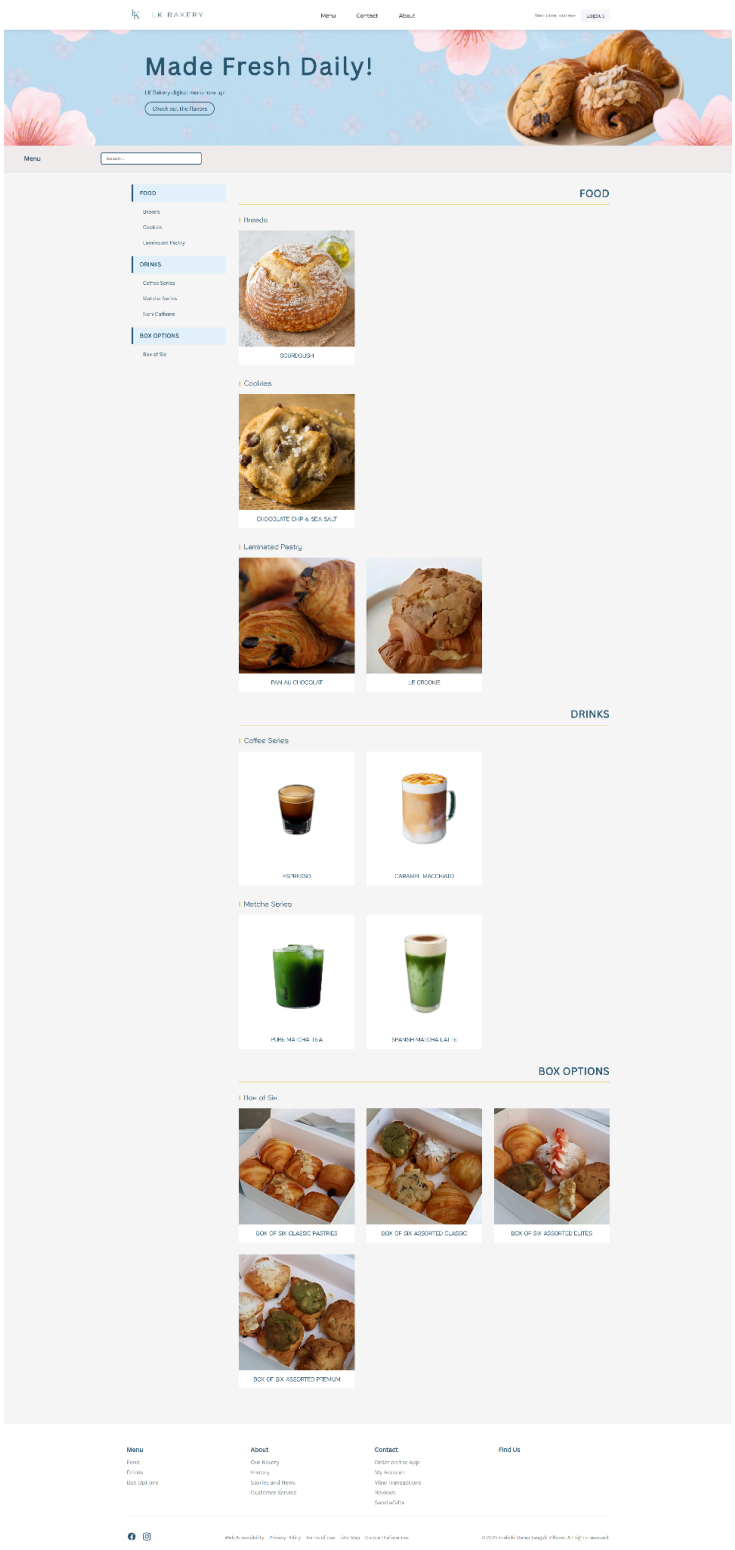



Figure 2.1 Menu Page

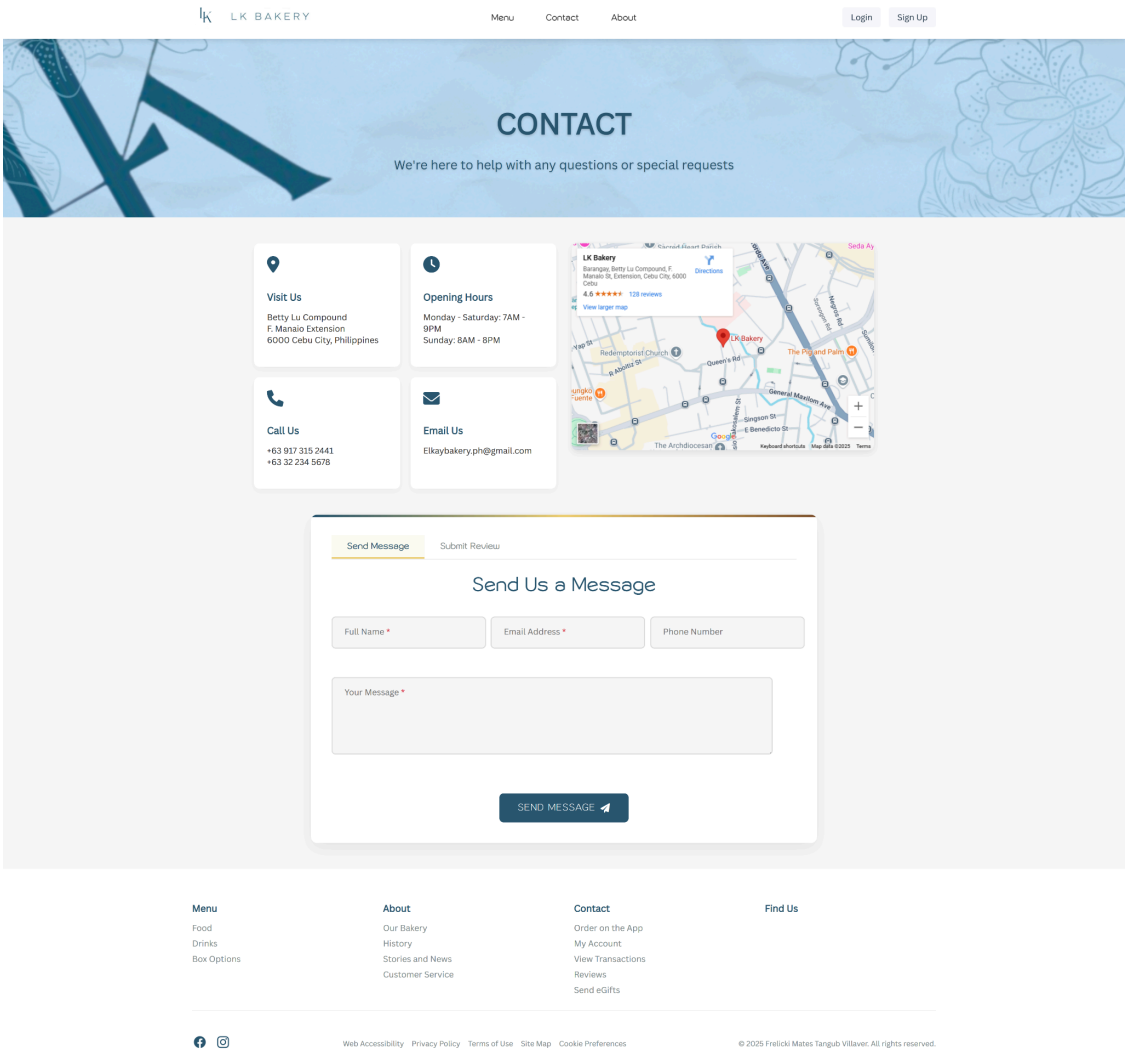



Figure 2.2 Contact Page

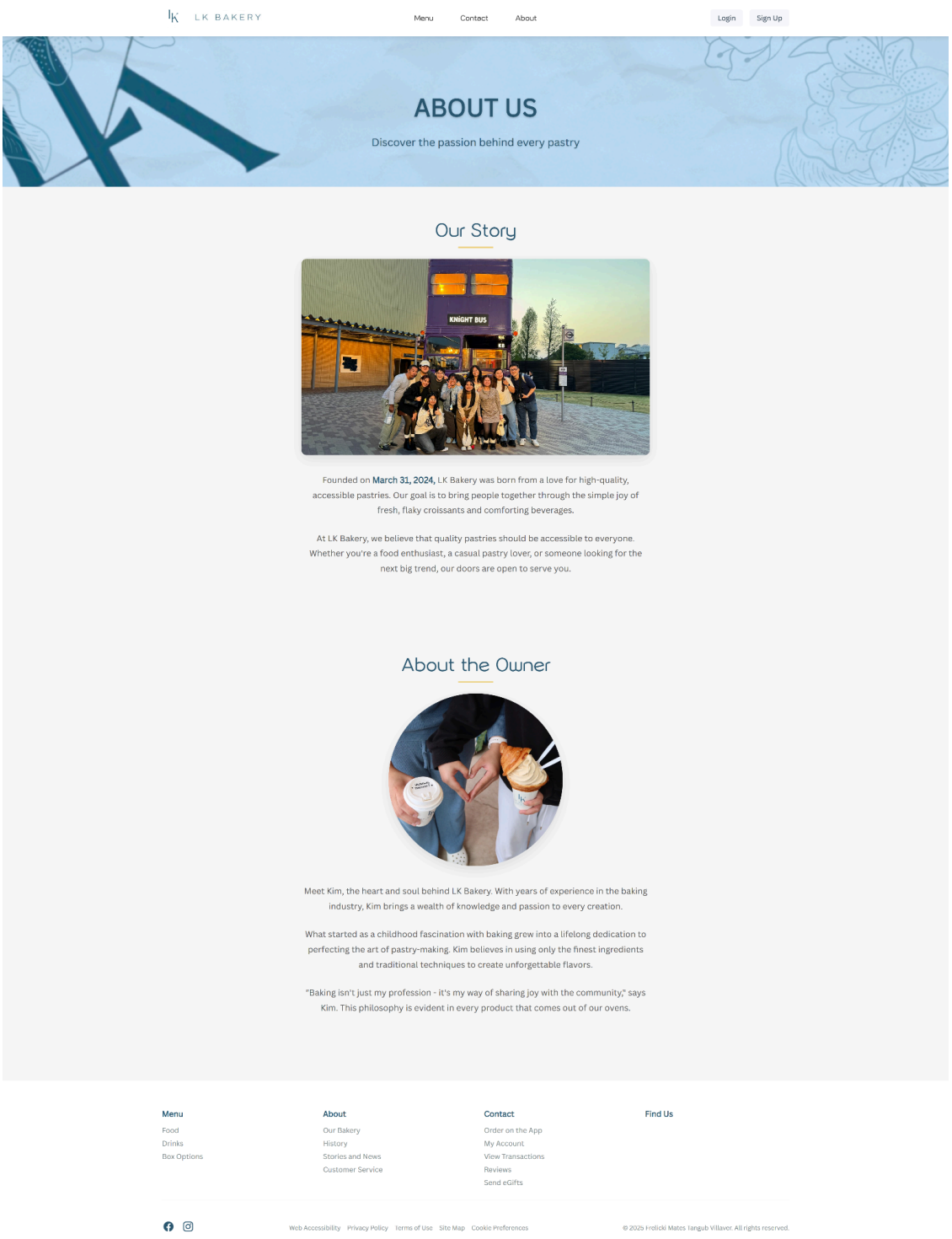



Figure 2.3 About Page

## Backend

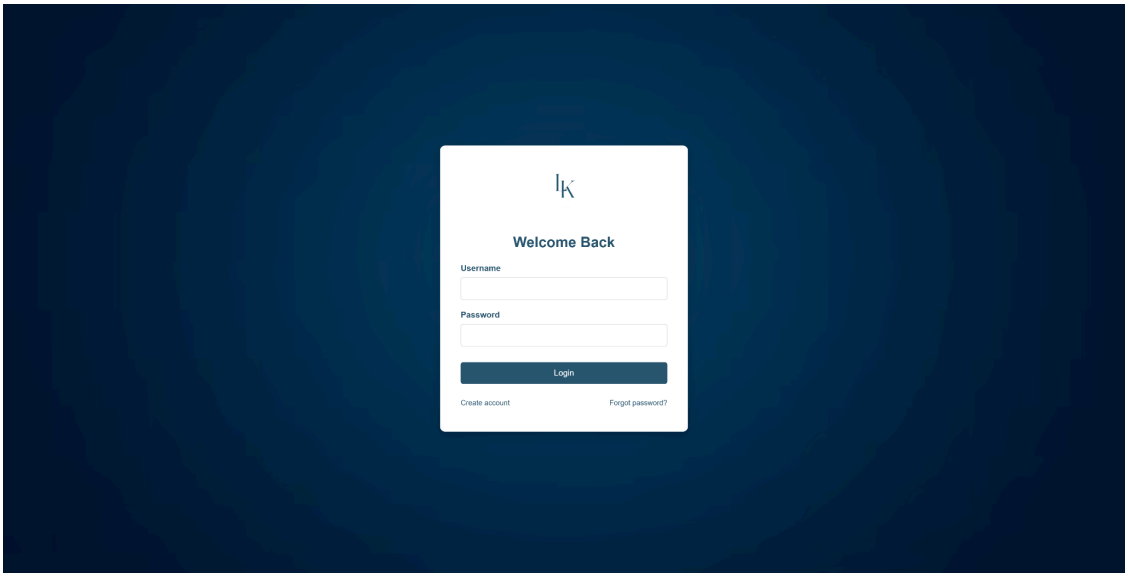



Figure 3.0 Log In Page

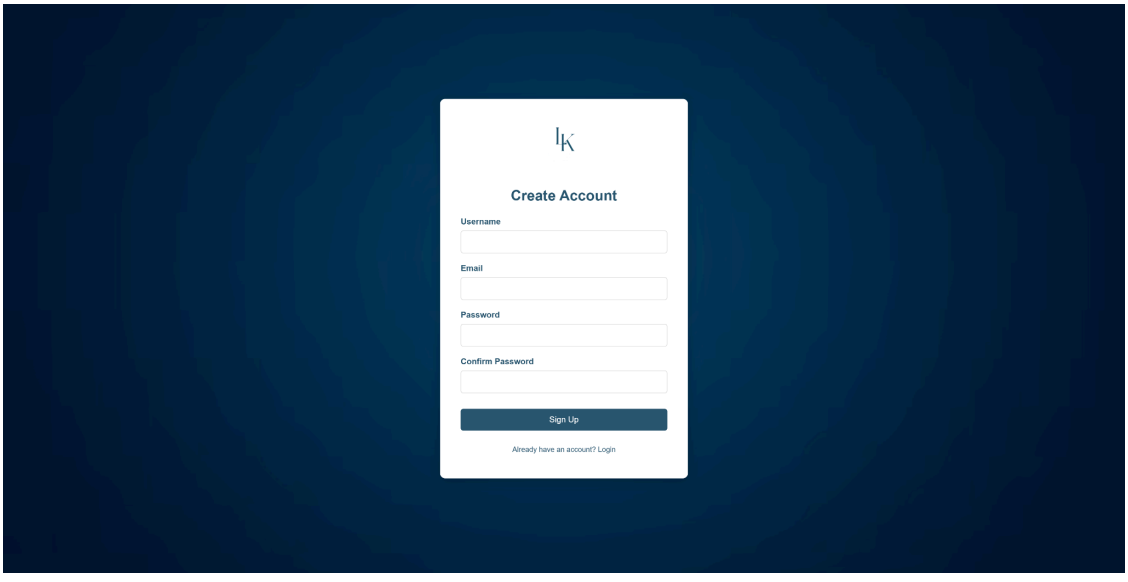



Figure 3.1 Sign Up Page

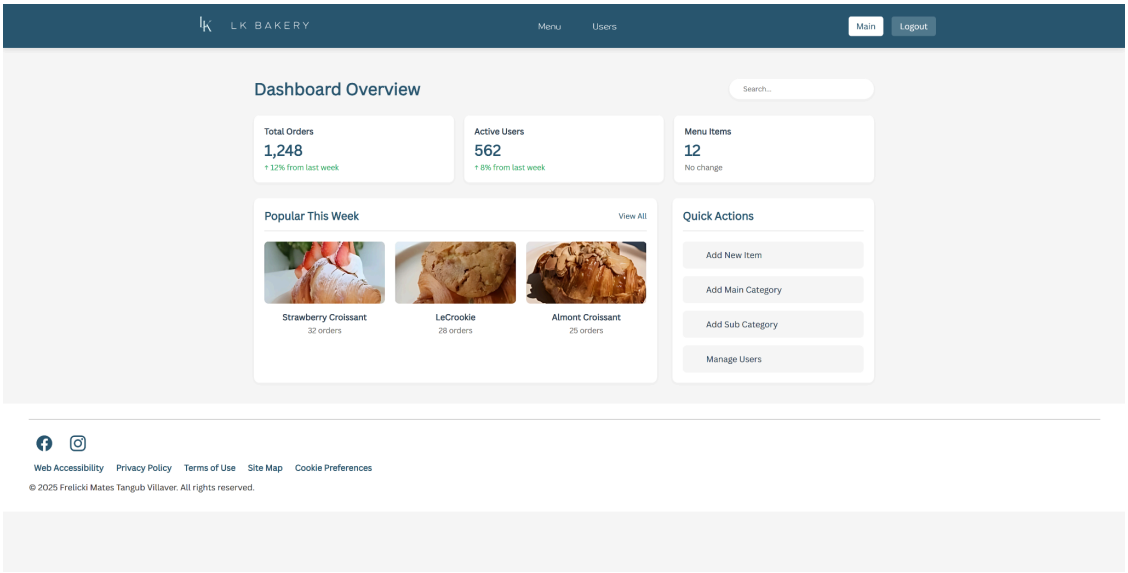



Figure 3.2 Dashboard Page

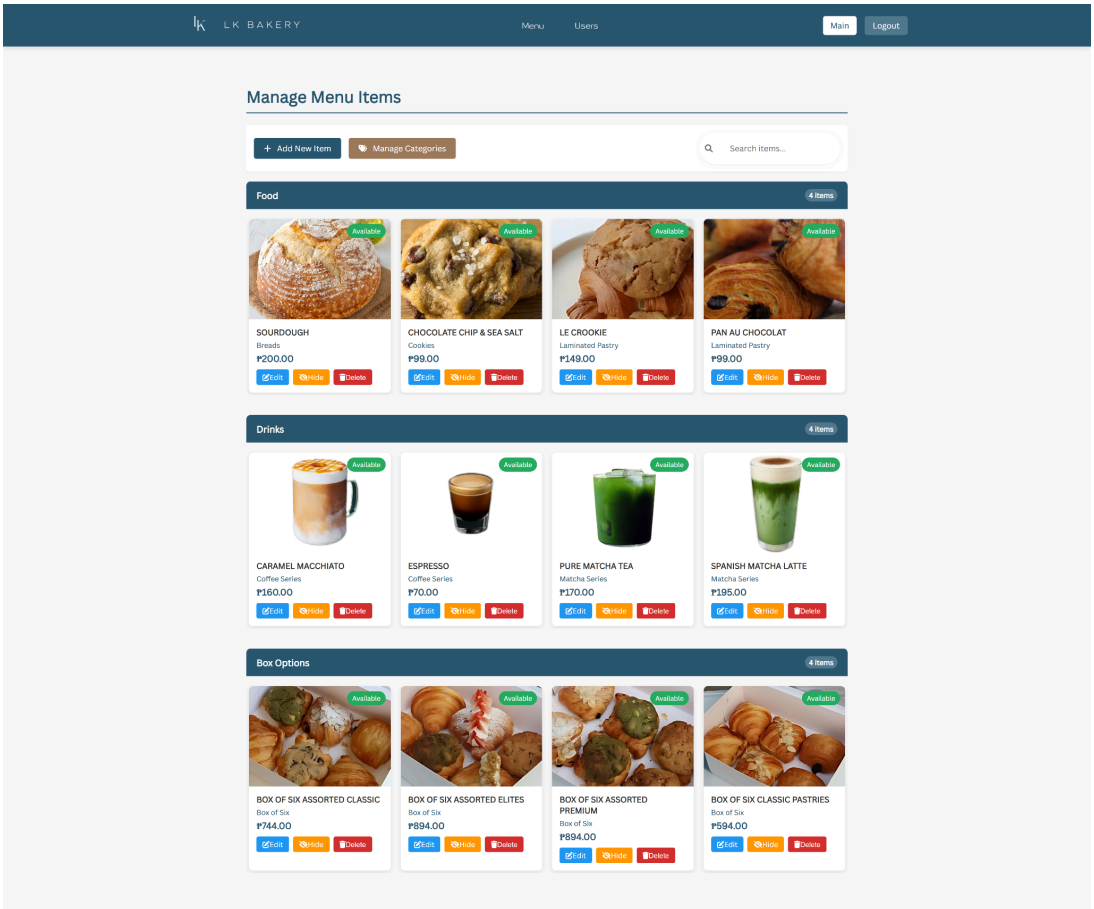



Figure 3.3 Manage Menu Page

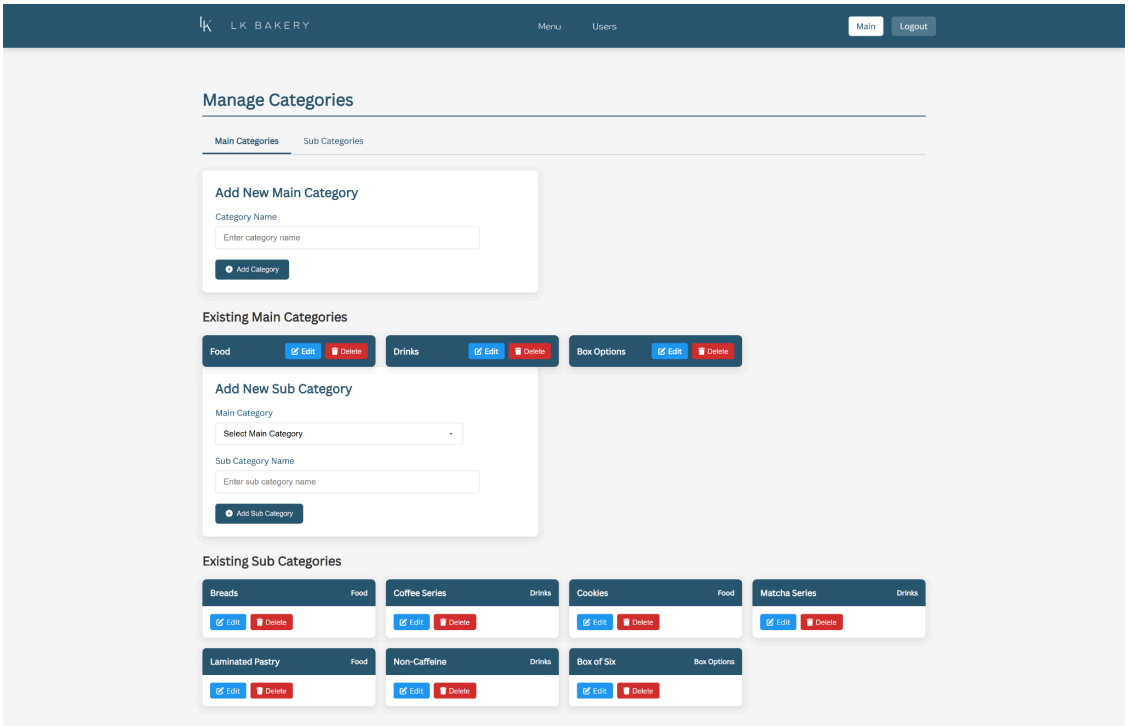



Figure 3.4 Manage Main Categories Page

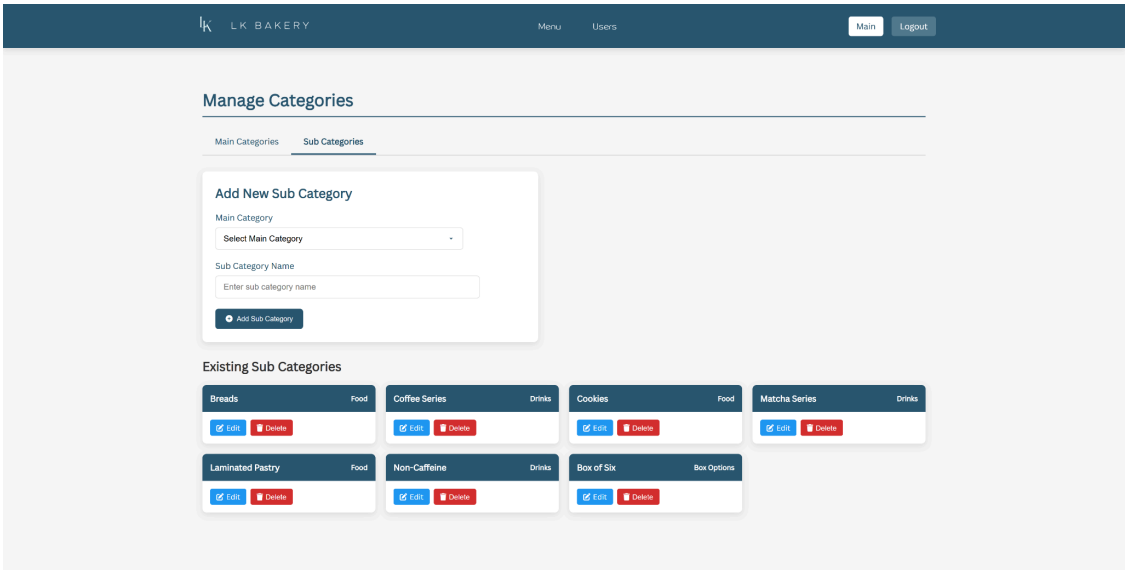



Figure 3.5 Manage Sub Categories Page

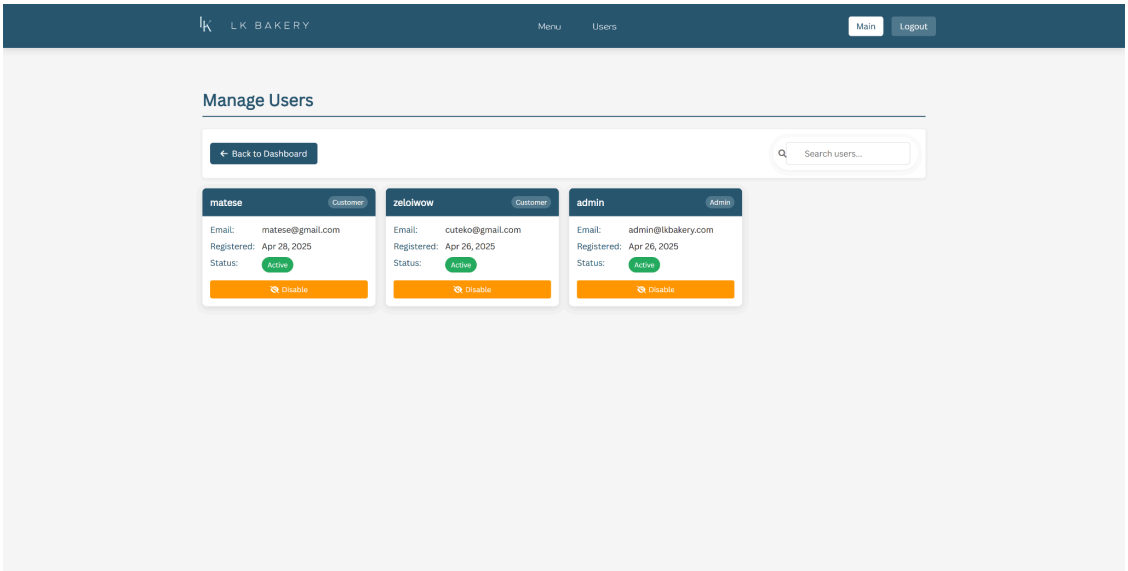



Figure 3.6 Manage Users Page

# Appendices

## Appendix A

**Client Interview Summary – LK Bakery**

*Conducted for a Web Development Project Proposal*

Below are some questions organized into sections to help us align the project with your goals. Please note, we want to ensure we respect the privacy of your business, so feel free to leave any questions unanswered if they may compromise the confidentiality of your business.

1. **About LK Bakery**
   - ***a.*** *What products or services do you offer?*
     - Laminated pastries and complimentary drinks such as coffee, matcha, and tea.
   - ***b.*** *How long has your business been operating?*
     - Our operations started on March 31, 2024
   - ***c.*** *What is your target market or customer base?*
     - People aged 18 - 35
     - Food enthusiasts
     - Clout chasers
   - ***d.*** *What are your long-term goals for LK Bakery?*
     - Broaden our customer base and make quality pastries accessible to all
     - Being able to reach customers outside Cebu. Like selling through tiktok, shopee, etc.

2. **Goals**

   a. *What are the primary goals you want to achieve with this website project?*

      - To educate our customers about the kinds of ingredients we use and emphasize the story behind LK Bakery and the mission of the bakery, which is to make quality pastries more accessible for everyone.

   b. *Are there any specific outcomes or improvements you are hoping to see?*

      - Increase in online orders

   c. *How do you envision this website helping LK Bakery grow or better serve your customers?*

      - The website would be more of an introduction to LK Bakery to the world.

3. **Business process**

   a. *How do you currently handle customer orders?*

      - Through walk-ins, online orders, and Grab

4. **Proposed Site Process**

   a. *What features or functionalities would you like to see on the website?*

      - Location
      - Contact info
      - About the owner
      - Mission and vision
      - Our products & descriptions
      - support

   b. *How could the website improve your current process?*

      - It will drive in more customers from different places around the world

   c. *Are there any specific tasks you would like to prioritize on the website?*

      - none

‘

5. **Site's Scope & Limitations**

   ***a.*** *Are there any additional features you would like to see in the future?*

      - None I have in mind

   ***b.*** *Are there any specific limitations or constraints that we should consider?*

      - None I have in mind

6. **Suggestions**

      - Review Section

## Appendix B

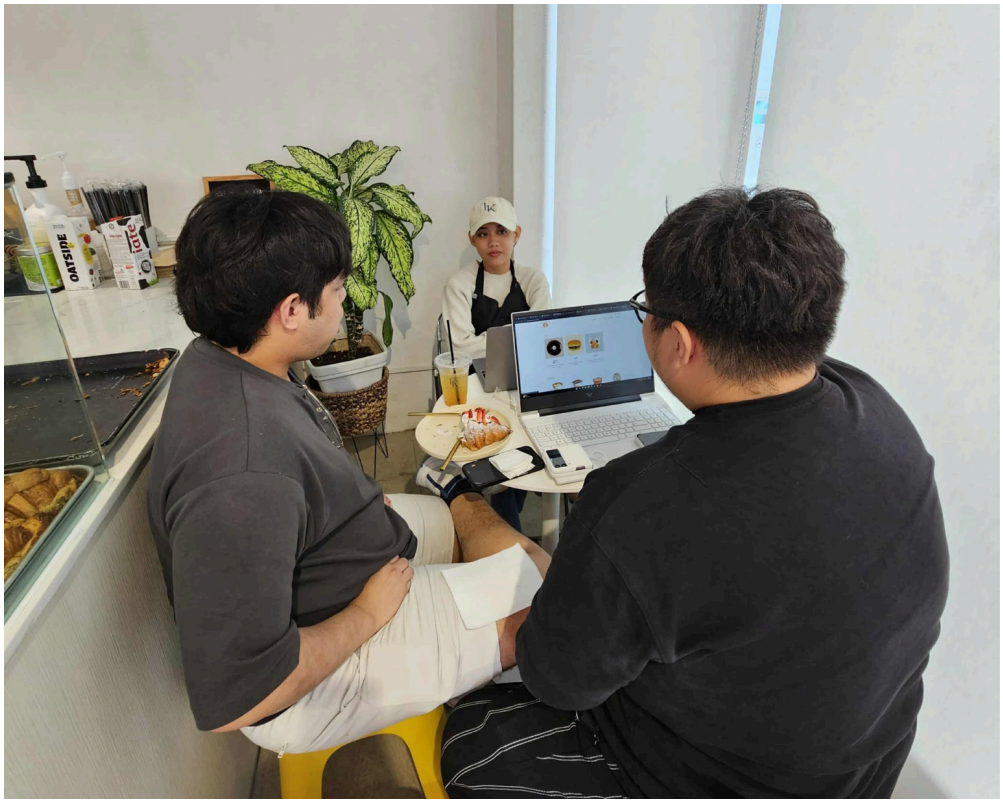



Figure 4.0 Photo of the Interview